ActInf Textbook Group ~ Cohort 4 ~ Meeting 19 (Chapter 9 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | *Cohort 4 ~ Meeting 19 (Chapter 9 part 1)* | *56:56*

All right. It's cohort four, October 31st, 23, and we're at our first discussion of chapter nine.

So today we can totally look at questions and look over the chapter, but is there anything that anybody wants to bring up about chapter nine?

Thank you.

Before we kind of look over the structure of the chapter and look over the questions.

Okay.

We'll just hop into the structure. A good overview seems nice on this one.

Cool. Yeah. And Esmail had mentioned epistemology. So talk about leading with a epistemological assertion. The models described in this book are only useful if they can answer scientific questions.

Is that the only way a model can be useful? Or is this such a broad sentiment that everything can be cast as a scientific question?

So even something of purely epistemological or epistemic status could be under this scope?

Anyways, in this chapter, we focus on the ways in which active inference can be applied to understanding empirical data.

So that's kind of the big plot twist in chapter nine is that up till here, everything has been very much like proposed as sort of intrinsically coherent.

Like the rat in the T-maze or other calculations.

They're all kind of proposed as just thought experiments more than anything else.

Although there's a lot of citations to the empirical work like in chapter five.

Here, though, they're going to go into detail about how you interface the generative model with empirical data.

And so in that way, it kind of complements chapter six, because here we're actually going to take that generative model and connect it with data.

Our general goal is to recover the parameters of the generative model that a subject's brain uses to produce behavior.

The subjective model.

So we'll see this graphically soon.

But just to be clear, to recover the parameters, a synonym there would be like to identify plausible parameters.

So if we knew that the height of children in the classroom was distributed according to a normal distribution.

Then we have the data, which is kind of distribution free.

It's just a list of numbers.

And then the task of chapter nine is basically juxtapose the structure of that model with the data and recover the parameters.

Okay, section 9.2 goes into this metabasean perspective or metabasean methodology.

They bring up two related reasons for fitting a computational model to observe behavior.

The first is to recover parameters to give an account of a single individual from a broader population or comparing two groups or two populations or more against each other.

So that is taking a fixed model structure and then multiple samples of data.

It could be one person on multiple days or multiple people on multiple days or whatever the design of the experiment is.

And then for a fixed model structure, identify something useful about that data, like about one group versus another.

The second reason is to compare alternative hypotheses expressed as models that represent different explanations for behavioral phenomena.

So here, instead of preconditioning on one model, we're actually going to use the data to interrogate a portfolio of models.

So we have performance of multiple individuals on multiple days.

So then we might say, okay, well, the simplest model is all individuals are the same, drawing from the same distribution, all days draw from the same distribution.

Then we have a model where individuals don't differ, but days differ.

Then we have individuals differing, but not days.

And then we have, there's an individual by day interaction.

So this is familiar to anyone who's basically done any kind of statistical modeling, which is that adding more parameters to a model, essentially, or almost by definition, always increases the accuracy of the model on the training data.

That doesn't mean it increases the accuracy in and out of sample.

So that's like the training and test relationship of overfitting.

But adding parameters basically always gets you to fit more data or fit the data more closely.

However, the return on investment per parameter can diminish.

And so that is why in Bayesian statistics, people use methods like the IKK information criterion, AIC, or the Bayesian information criterion, BIC, to identify, well, where are we adding parameters?

Where's this kind of like knee or this special inflection point where, yeah, of course we can still add more parameters and get more fit,

but we're like basically at a nice optimal tradeoff between adding parameters and fitting the data, which is exactly what variational free energy is.

So those are two reasons.

One is to take a model, identify something interesting, like a pattern in the data.

The other, which is complementary, is to use data to identify or differentiate hypotheses in the world.

So yeah, actually having the day factor and the individual factor, those add value, but having an individual by day, we dilute too much.

Just like raise your hand if you have any thoughts, if you want to add anything.

So this is a pretty standard statistical framing.

Here is where we're going to revisit some of the key concepts from earlier in the book, like the generative model of chapter four, discrete time model, chapter seven, all of this.

And now we're going to kind of put it in a laboratory context.

So we have the subject of a behavioral experiment.

So let's not worry too much about like objectivity of truth and subjectivity of experience, but let's just think about a behavioral study.

So the subject of the behavioral study is the one whom we are developing the cognitive model of.

Their cognitive model, their generative model might be about the laboratory, like the maze.

However, the subject model is what we've been discussing in basically all of our kind of agent-centric cognitive modeling discussions.

And then in chapter nine, here we're just going to pop out and add another dashed ring and a solid box around that to represent the fact that it's kind of like now we have the lab and the subject.

Here's the experimental stimuli going into the black box.

Here's the observed behavior objectively coming out, or it could be said empirically coming out.

So this kind of gives a whole enclosed or a closed loop representation of like the person in the fMRI machine.

And then here's the experimental stimuli being provided to them.

Could be a fixed sequence.

It could be drawn from a distribution of stimuli.

And then there's the behavioral outcome.

So even the setting of the behavioral observational moment, the ethological setting, this is also amenable to Bayesian statistics.

So we could talk about optimal design of experimental parameters.

We could talk about optimal delivery of different stimuli.

We could talk about the information content for different sequences of behavior.

We could do a statistical power analysis or simulation.

Should we observe a few individuals for a long time?

A lot of individuals for a short time.

These are like very empirical laboratory statistical considerations that will be familiar to anyone.

Basically doing a behavioral study or really just any other study, even if it's not like squirrel behavior.

But if you have a limited amount of resources or time or cameras or whatever it is in the lab, you're going to find yourself basically in a setting like this.

And so that's why this is a really nice figure and chapter.

Because it steps outside of just the kind of abstracted generative model.

And now it's like, okay, now we're going to build the setting around that.

And so it's kind of here's outside the building.

But now it's like the laboratories, like the Markov blanket around the rat in the maze.

Which is to say, if you control the laboratory, if it's a controlled experiment, then what's happening outside of the laboratory shouldn't matter.

Should be conditionally independent with respect to the subjective model.

So this is kind of compatible with a lot of Bayesian epistemology in empirical sciences.

Okay.

Any thoughts or questions on that?

Or we'll continue with end of 9.2, then 9.3.

It's called meta-Bayesian, or I don't know, maybe you could say epi-Bayesian or something, double Bayesian, because of this wrapping with a Bayesian statistician around the Bayesian generative model.

And then they even say, we're using Bayesian procedures twice.

First to evaluate the subject's posterior beliefs about action.

So posterior means posterior to the observation.

So we're going to flash the red light.

Then we observe an action.

And then we're going to do parameter recovery about the subject's posterior beliefs on action.

Posterior to the observation of the red light.

That's the subjective model.

And secondly, to evaluate our posterior beliefs about the unknown priors that we have.

So it's just a purely Bayesian approach to studying the total setting of the behavioral observation.

Otherwise, you'd have this kind of free-floating Bayesian model of the subject.

But it would be like a view from nowhere.

Because you wouldn't have completed the meta-Bayesian setup.

Okay.

That's 9.2.

9.3 is going to connect us with a statistical method that's used in a lot of statistical approximation algorithms.

And it's discussed in SPM a lot as well.

So variational Laplace may be used for more generic likelihood functions than encountered earlier, which were defined as Gaussian.

So variational Laplace.

We won't go through all the details here.

But usually what the Laplace approximation is going to do is it's going to find the single highest point on the distribution, the single maximum likelihood solution, which is why the likelihood calculation from earlier matters.

And then going to fit an upside-down parabola.

So if the distribution has a central tendency, the variational Laplace can do very well.

If the distribution has fat tails or has a multimodal tendency, variational Laplace will not do well if it's Gaussian.

Kind of a technical subsection.

But it's possible to tractably optimize the variational Laplace.

So it's going to be misleading in some settings, but it's going to be super useful in some settings.

But either way, it's very calculable.

So it's used for a lot of statistical optimizations.

Okay.

So a common critique and valid point that people often bring up about Bayesian statistics is, well, don't you just get out what you put it?

Like, if your prior is such and such, then don't you just recover your prior in the posterior?

And there's multiple ways to address that.

One of them is if you start with multiple different priors, so from the same family, parameterized differently, or from different families, and you recover your prior,

then you actually can empirically say that there isn't information in your data.

But to the extent that those diverse priors converge on a similar posterior, you can empirically say that there is signal in the data.

It's not binary, but those are different heuristics that are used.

So one approach to kind of get around or to address this reviewer type comment that like, oh, well, Bayesian statistics just returns your priors is to try different priors.

And there's also another approach, which is called parametric empirical base.

So it's called empirical because we're taking empirical data and using that to initiate the cycle of optimization.

So, for example, if we were going to be doing a Bayesian model of the height of the children in the classroom,

one approach would be like, we know that people range from one feet to 10 feet tall.

So we're going to do six feet plus or minus five as a Gaussian.

Another approach would be, wow, we really don't know.

It could be between zero and a thousand feet or something like that.

Like just try to do this sort of like purposeful open-endedness, but that might wash out true signal in the data.

Like even if a bunch of people were the same exact height, you've only shrunk your distribution now to between zero and 500 feet tall.

So what's the way to get around that?

Well, one is trying multiple prior sets, but another one is to take the first batch of empirical data,

parameterize a Gaussian based upon that, and then use that Gaussian or an over-dispersed form of it as your prior for subsequent analyses.

This discussion on design matrix, so representing the experimental design as a matrix,

like if we had five people on multiple different days, you have like data point one,

and then in person A, they'll have a one, and then all zeros for the other columns.

So representing the design matrix of the experiment as a matrix,

and then putting that within a generalized regression framework

is discussed more in the SPM textbook than in this textbook.

So this is kind of another technical subsection

talking about how you can use empirical

guidance on initial prior sets

in a generalized linear modeling setting

with experiments summarized by their design matrix

to do a regression on model coefficients

so you can do statistics on those coefficients,

like testing whether the slope is different than zero.

So to go back to the earlier example

where we had like multiple individuals on multiple days,

you have the simplest model,
no effective individual, no effective day.
And then you have model two, effective individual,
model three, effective day,
model four, effective both,
or model three, effective both.
And then you can test
the relative likelihoods of these different models,
and then for whichever one is the best selected model,
then you can say,
okay, the coefficient was 0.1 plus or minus 0.5, 0.05.
So the slope was two standard deviations away from flat.
So at this alpha level,
it was a statistically significant slope
for the best selected model.
So it's kind of speeding rapidly
through a few different parts of statistics,
but that's basically what is happening
when you just smash together the design matrix
and the experimental coefficients.
You're basically doing a regression,
and then you're calculating essentially a p-value
based upon the slope of the regressions
conditioned on the design matrix.
Okay.
Okay.
So actually, just,
I'll just raise your hand or unmute her
if you want to add anything,
but just kind of going over this chapter.
9.5 is a lot like chapter six.
Chapter six was the recipe
for making a generative model.
9.5 is the instructions for model-based analysis.
So 9.5 is almost like a coda to six.
So the steps are summarized in figure nine, two.
First, collect data,
or you might collect data later,

and you might do all these other steps earlier,
like if you're going to pre-register the study
or do a simulation or a statistical power calculation.
So you don't have to collect the data first.
Again, just like chapter six,
this is kind of like more like stuff that has to be done,
not really a list that must be done in order.
However, at the very least,
you'd want to know like what sensory input
is available to the subject
and what data you're getting out.
Sensory data for the subject,
what data are coming out?
As a kind of side note,
many studies have been waylaid
because there was sensory data
that wasn't intended
that was getting to the subject.
So I've heard examples ranging from,
it was a study of circadian behavior
and there was construction in the building.
So at certain times a day
or during certain cohorts or replicates,
there was sound or just in a animal facility,
different animal handlers
or different light schedules
or different temperature,
different air conditioning,
forest fires that we had in California,
like the sensory inputs available to the subject
doesn't mean just the ones that you want.
And so that can disrupt a lot of the statistics.
But this is basically like saying,
what data are you going to get
from what kind of subject?
And then what type of tests
are you interested in doing on that data?
There's always exploratory data analysis,

EDA,
also criticized as being kind of like
a fishing expedition.
So sometimes people go into a study
with really specific hypotheses,
other times more open-ended.
Two,
formulate a POMDP model
if you're using a discrete time setting,
which is kind of how they're going to do it.
They're not going to template this
with a continuous time model.
So we get a fully specified
but not solved POMDP.
So structurally described
but not parameterized POMDP.
Specify a likelihood function.
We can look a little bit more
at SPM, MDP, VBX
to try to see exactly
what degrees of freedom are here.
but my understanding is
this is not something that needs to be designed
on like a per-experiment level.
specify prior beliefs.
These are the priors
that are being proposed
for that meta-Beyesian setting.
So if it's kind of a,
if it's,
there's some priors
that are centered on zero.
Other times
you have a prior distribution
over a distribution.
So it's like,
it's bounded by zero and one.
at which point you could use uniform

or some other
proper distribution.
Solve for posterior probability.
Standard,
standard inference
scheme.
Again,
we could look at this
routine
but this just takes in
the above
information
and makes a calculation.
So the output of steps
one through five
is going to be
these estimates
of different parameters
of the model.
Then
you can do
the group level
analysis.
We can look at SPM DCM.
This is the dynamic causal modeling.
So that's one
approach
that's discussed
more in SPM
or
you can basically
use any
statistical method.
Canonical variance
analysis,
CVA,
similar to the
independent contrasts,

ICA,
or principal components
analysis,
PCA,
or just ANOVA,
any number of
statistical analyses.
So here's the summary.
We have the data.
We have the
POMDP
and the way
that it's going to get
calculated.
all of this
gets brought together.
So
U
tilde
always means
through time.
And here's U
tilde
the action
as selected,
as observed
behavior.
behavior.
So that's what
we're getting
from the data
is the behavior.
Survey data,
movement data,
eye gaze data.
And then here
is going to be,
we're going to get

a likelihood
function
for saying,
what's the likelihood
of given behavior
conditioned upon
the parameters,
big theta,
all the parameters,
the sequence of observations
received,
and the generative model,
M.
So that is kind of
what the subject
is doing,
is it's doing
this likelihood
calculation.
What's the
likeliest
behavioral sequence
given
basically
all the parameters
accessible to me,
the cognitive model,
as well as the observations.
and then
we invert
that.
And that's
now happening
on the scientist's
side.
So
we're going to
invert so that

we can recover
the parameters,
big theta,
conditioned upon
the model,
the observations
of the subject,
and the action
of the subject.
That
is
proportional
to
so
it might
be off
by a
multiplier,
but it's
monotonically
proportional
to
a
factorized
disassembly,
which is
the parameters
given the
model,
which is
something that
can be
taken off,
multiplied
by
here
exactly
what we

have here,
P
of U
tilde
conditioned
upon
theta
observations
in M.
Now,
we can
specify
our
prior
beliefs
on this
left
side,
because
this is
basically
our
beliefs
about
theta
given
the
model.
And again,
we would
test a
range
of
these
and or
use
parametric
empirical

base
to generate
empirically
the ground
of beliefs.
But with
this
inverted
model,
we can
then do
statistics
on
calculation
of data.
We have
X,
the design
matrix,
beta,
which is a
variable with
the regression
coefficients,
and a
noise.
So it's
kind of like
Y equals
MX plus
B plus
noise term,
except it's
like a
mega regression
on a
much more
sophisticated

kind of
data.
But in
the end,
here's those
five groups.
And then
here's this
one's
parameter
range was
0.2
plus or
minus 1.
This one
was 5
plus or
minus 1.
This was
4 plus
or minus
1.
And this
is like
the kind
of figure
that you'd
get in
the paper,
where there
would be
like a
bar A
saying,
oh,
these are
not
significantly

different.
Bar B,
these two
are not
significantly
different.
And then
an asterisk
saying
these two,
you know,
individual
1,
2,
and 3
were
significantly
different
than 4
and 5
at this
p-value.
So
chapter 6
builds us
up to
constructing
the generative
model.
If you
have
empirical
data
already
at hand
or if
you intend
to collect

empirical
data and
you already
know what
structure
it's going
to be,
you kind
of design
this
complementary
relationship
between
the
observations
that the
generative
model is
going to
get
and then
the actions
that the
generative
model takes,
which are
going to be
observations
for the
behavioral
scientist.
And then
if the
behavioral
scientist
can specify
this
likelihood

model
and a
prior
belief
distribution,
we can
invert
what it
is that
the
subjective
cognitive
model is
doing
to
recover
differences
or patterns
amongst
parameters
for different
individuals or
times or
contexts.
tests.
So this
takes us
from a
chapter
six,
which is
just how
to build
the
generative
model and
lays out the
big steps

for how we
would then
get to the
figure for
the paper
or to
test,
okay,
here's five
different doses
of a
drug.
And then
on the
y-axis is
the
attention
parameter,
variance on
this,
which is
we're going
to interpret
that as
attention,
and here's
how it
differs across
doses of
a drug.
So it's
a lot
of
statistics
here being
very,
very quickly
recited

in a
Bayesian
cognitive
behavioral
modeling
setting.
Could you
repeat the
example with
the drug
dosage again?
Yeah.
So let's
just say we
had five
different
doses of
drug or
control and
you know,
low,
medium,
high,
and very
high.
And then
each of
those,
we're just
going to
do,
you know,
of course,
you can
have N
individuals on
N days
with N

doses.
That'd be
like a
fully replicated
experiment,
but you
have some
subset.
So we
have five
groups,
and then
we have
30
individuals
per group
rats,
so that
we don't
need to
talk to
the IRB.
We probably
still do.
So then
you have
the rats
do these
different
behaviors.
Now,
if you
just wanted
to do
statistics
on their
behavior,
you wouldn't

even need
a cognitive
model.
So if all
you were
going to
do was
going to
do,
like,
I'm
going to
do the
average
percentage
of time
moving
between
the five
groups,
well,
then you
could just
read that
off from
the data
and just
say,
okay,
it was
10%,
15%,
this and
that.
So you
could have
a little
scatter plot

and then
you could
say,
do these
groups
statistically
differ
in the
percentage
of time
moving?
So that
would just
be like
literally
from the
data
to step
six.
However,
you might
also have
a cognitive
model.
And so
there you
would take
that data
and then
see the
actions
of that,
the data,
which are
the actions
of the
subject,
and then

utilize this
constructed
cognitive
model to
do inference
on some
cognitive
parameter of
these five
groups.
So not
simply
descriptive
statistics on
the action
output.
And then
these
scores
are
referring to
some
proposed
cognitive
parameter.
What's
like an
example of
a cognitive
parameter?
I'm very
interested in
like, I
like, yeah,
I like that.
I just want
like an
example.

Yeah.
That's a good
question.
Let's look
to some
of the work
of Ryan
Smith.
Okay.
Let's see
if this
will have
a nice
generative
model.
Some of
these will.
Well,
perfect.
Perfect.
So here
we
see our
classic
downstairs
bit,
figure 4.3
downstairs.
The
empirical data
are going
to be
the
heart
beating
empirically.
And then
the hidden

state is
going to
be, I
believe, in
this situation,
the
interoception
of the
heartbeat.
And then
A, which
of course
maps between
observations and
hidden states,
if A were
the identity
matrix, this
would be a
fully observable
setting.
But of
course, A
isn't exactly
the identity
matrix, and
in fact,
they're
interesting in
the precision.
So then
they can
parameterize
this model,
and you
could say,
and then
you can

test whether
the precision,
the
interoceptive
awareness,
differs
amongst
individuals,
or between
individuals who
have this or
that diagnosis.
Let's see if
there are other
models.
Here's another
one.
This one has
to do with
gastrointestinal
perception.
Again, we
see the
downstairs.
So then
what would be
the next
step?
What would
be the way
to integrate
action?
Well, we
know action
intervenes in
the B
matrix.
So now

if you had
a descending
action
condition,
and then
you could
say,
okay, now
do or
don't try
to pay
attention
to your
gut.
And then
you could
do statistics
on the
efficacy of
the B
matrix.
statistics.
But now
these are
making
larger and
larger
experiments,
and that's
where you
get back
to the
resource
limitations
in
laboratory
and
stuff

like
that.
But,
yeah,
then you
could say,
well, now I'm
interested in
this summary
statistic
describing
the difference
between the
average
precision when
we do and
don't tell
them to do
this,
and there's
some
individuals
where there's
no difference,
like telling
them to pay
attention or
not didn't
differ their
precision,
and there
were some
individuals
where telling
them did
alter their
precision.
But, of

course, those
precision estimates
are features of
a statistical
model, which
is why we
had to
construct this.
Those precision
estimates and
the statistics
about how
those precision
estimates
differ,
those are not
features of
simply the
heart rate.
But this
is a big,
apparatus to
bring out.
So, if you
just are
actually
interested in
the descriptive
statistics,
if your
question can
be addressed
with description
of behavior,
you literally
don't need a
cognitive model.
This is being

brought out when
you are trying
to make
inference about
something that's
not directly
objective.
negative.
Then they
reference these
two papers
that have
slightly
different ways
of looking
at this
eye
saccade
situation.
again to
this kind
of Bayesian
epistemology
and also
like sociology
of
neurodiversity,
really.
We hear so
much rational,
irrational,
irrational,
non-rational,
all this.
and
active
inference or
just Bayesian

epistemology
has kind
of like a
simple and
a deflationary
but also
kind of a
maddening
answer to
that,
which is
all the
generative
models are
on their
path of
least action.
So,
those paths
may differ,
but the
paths are
all alike
in that
they're all
paths of
least action.
So,
it's all
Bayes
optimal.
One
person might
be engaging
in a
repetitive
behavior.
Another

person might
be ruminating.
Another
might be
having
anxiety.
Could be
human,
could be
non-human,
but those
are all
paths of
least action.
So,
obviously,
there's a lot
to say
about this,
but that
gives a
little bit
of a
space
between
like
disorder
implies that
there's like
an ordered
and a
disordered
but here
it's saying
inference is
working fine
but on a
flawed

generative
model.
But,
again,
you could
even go
further than
that and
say,
well,
there's
nothing
flawed
with it.
Like,
so,
but that's
the kind
of way
that it
takes
that
conversation.
Table
9.1
reviews a
bunch of
settings
where these
kinds of
parameters
have been
done.
addiction,
impulsivity,
compulsivity,
delusion,
hallucination,

interpersonal
personality
disorder,
oculomotor
syndromes,
pharmacotherapy,
prefrontal
syndromes.
That seems
pretty general.
Not sure
what that
one is.
Visual
neglect.
Disorders
of interoceptive
inference,
and that's
where the
Ryan Smith
work
especially
is relevant.
that's the
summary.
So,
it's kind
of an
interesting
chapter
with a
few of
these
mosaic
sections
beginning
with a

large
scoping
frame
on
cognitive
behavioral
modeling,
then
having
these
two
kind
of
technical
subsections,
!
then
going into
a chapter
six like
interlude
about how
this work
is empirically
deployed,
providing
two examples
of generative
models in the
eye-tracking
setting,
going into
this kind
of like
social
definitions
of
pathology

and
empirical
work on
pathology
space
summary.
people who are
interested in
this like
design matrix
or just
like
experimental
statistics,
SPM
textbook is
really the
place to
look.
Another
kind of
interesting
note is
the
methods for
active
inference are
very well
developed in
the SPM
toolkit in
MATLAB.
And so
for
neuroimaging
and those
kinds of
studies,

there's a
ton of
empirical
work.
However,
most people
who are in
data science
today are
not using
MATLAB,
using
Python,
and PyMDP,
leading Python
active
inference
package,
is only
now beginning
to include
these
empirical
going from
data to
the model
type
model
inversion
tools.
those do
exist in
the
Julia
Rx
and
Fur
world,

which is
one reason
I believe
why such
a common
kind of
paper today
is like
we built
a generative
model,
we
synthesized
some data,
we simulated
data,
because going
from the
data back
to the
model is
actually not
so easy
today for
a full
active
inference
model.
But as
that becomes
more
plausible,
there's
going to be
a whole
new
forest of
low-hanging

fruit for
those who
choose to
forage,
because this
data could
be any
data.
So this
could be
click data,
it could
be ant
foraging
data,
I mean,
these are
just regular
data sets
that people
have already
analyzed.
and then
it will be
a purely
empirical
question,
what
cognitive
modeling
can add
in
comparison
with purely
descriptive
statistical
methods.
So as

that
capacity
becomes
unlocked
in the
active
inference
ecosystem,
it won't
always be
like
situation
generative
model
simulation
will be
able to
start with
empirical
data about
real systems,
economic
data,
psychological
data,
and so
on,
and then
already have
that while
we're having
the generative
model discussion,
that's going
to reduce
a lot of
degrees of
freedom on

the generative
model,
because we're
knowing that
it's going
to interface
with the
data a
certain
way.
So honestly,
it's like a
whole mode
of discourse
and work
in the
active
space that
we just
don't see.
But this
chapter lays
it out
because where
we do see
that mode
of work
is literally
Fristin et
all's work
in MATLAB
with the
SPM
package.
Is the
SPM
textbook,
is that

on the
website
for the
institute,
or is
that something
I should
just be
on the
month?
I'll put
it in the
chat.
I believe
that this
is the
most recent.
Yeah,
SPM 12
manual is
probably the
most recent.
this is
probably one
to look at.
The last
print version
is this
one from
2007.
So it
has some
typos.
It
is less
about like
action in
the loop

and the
control
theory.
However,
the
relationship
between
sensor
fusion
and
hidden
state
inference,
neuronal
dynamics
inference,
this is
absolutely
a slam
dunk.
So it's
very,
very,
and even
just like
even just
seeing the
kinds of
statistical
methods that
are in
play between
neuroimaging
observations
and then
just the
cognitive
modeling.

and then
having
action
kind of
come out
of the
neuronal
dynamics,
one,
I think
that that
is not
the only
learning
path,
but I
think that
that can
give a
more solid
basis for
the kind
of downstairs
part of
the model.
And then
also when
we're talking
about like
attention
as mental
action or
covert
action,
common
theme,
that's
basically

like saying,
well,
we have
this
descending
internal
action
that's
intervening
in
hidden
state
dynamics.
That,
so then
it's kind
of like,
but SPM
is like
the train
and then
now it's
like,
okay,
well,
action
intervening
in attention
is like,
you know,
I don't
know,
throwing
like little
throwing
something in
between the
spokes of

the train
wheel or
having a
hand on
the train
wheel,
but this
train
downstairs
is basically
statistically
summarized
in SPM.
So it
doesn't go
as heavy
into the
action,
embodiment,
all of
these topics
that we
like know
and love
as part
of the
discourse
today,
but I
think it's
a really
useful
prerequisite
or background
work.
So that's
different from
the manual.

That's
yeah,
I'm not
exactly sure
what the
overlap is
between the
manual and
the 2007
book.
We should
definitely use
the language
model to
find out.
I may or
may not
have a
PDF of
this
book.
Yeah,
and it
just helps
give us a
sense of
what
Friston
et al.
were working
on only
15 years
ago,
primarily.
And that
helps
contextualize
like, oh,

this whole
like physics
of cognitive
systems versus
just the
statistics of
neuroimaging,
but how
how nicely
they come
together.
And as
someone who's
never done
neuroimaging,
it was just
kind of cool
to like see
the kinds of
challenges that
they have.
And there's a
lot of
transferable
information.
Like the
fMRI bold
signal has
like a latency
of several
seconds.
So, you
know, it
takes two
seconds to
come on and
it takes 10
seconds to

fade off or
something.
So, then
if you're
timing
experimental
stimuli, the
stimuli can't
come in every
half a second.
But also, if
you wait five
minutes, you're
going to reduce
your statistical
power because
you only have a
fixed amount of
time in a
scanner.
So, like, these
are like very
empirical, very
pragmatic
aspects.
So, it
just makes me
think about
how in kind
of future
textbook groups
or future
internships or
whatever it
is, like
there's the
journey of
chapter six.

There's the
journey on
the low
road and
high road.
Then there's
the journey on
chapter six to
build a
generative model
for a
situation of
interest.
And then
there's the
journey of
doing an
experiment.
and of
how you
reckon with
these resource
constraints and
with data that
don't uniquely
identify models,
all these
different things.
So, there's a
lot of cool
things that
chapter nine
helps open
up.
Any just
thoughts or
ideas that
people are

having at
this point
in the
group?
Yeah, I
find this
chapter to
be just
incredibly
important as
far as
application of
active
inference goes.
And I
think it
was great
that you
kind of
directed us
to Ryan
Smith's work
and others.
I think
that there
are enough
general
principles
here for
this chapter
to be
useful.
And that
said,
actually being
able to
see like
direct

examples of
the application
in already
published
works seems
very, you
know, useful
in that we
actually have
something kind
of more
concrete to
get our
hands into.
Yeah.
So, here's
like interoceptive
precision.
So, this
is a, this
is a, they
did
parameter
recovery on
the A
matrix.
Then, you
can say, well,
what's the
correlation
coefficient
between this
coefficient we
identified,
IP, and
these
empirical
phenomena?

Reaction
time, reaction
time,
variability.
These are
just descriptive
statistics.
But on the
x-axis, the
rows are
precisions, a
difference in
the precision,
prior learning
rate, and
learning rate.
So, the
rows are
actually
cognitive
parameters.
And then, the
columns are
empirical
descriptive
statistics.
Blue is a
positive
correlation, and
red is a
negative
correlation.
correlation.
So, this
is immensely
transferable if
you think
about, well,

the observations
and the
summary statistics
are on the
blanket.
blanket.
So, if
you're just
trying to
draw conclusions
about just
the blanket,
you know,
there's a
factory.
It's making
blankets.
If you just
wanted the
average color
or the
average size
of the
blanket, you
could just
measure.
But then, if
you wanted to
do some
sort of
inference about
what was
happening in
the factory,
and then you
could do this
kind of
correlation

between
here's the
outcomes and
here's the
cognitive
parameters that
might lead to
those outcomes,
this starts to
really,
find some
interesting
possible
because you
and you
could also
look at the
correlations of
different cognitive
parameters.
And then here
it is with
the statistical
differences.
And you
say, well,
you know,
we don't
find a
statistically
significant
relationship
between
interoceptive
precision and
BMI.
Negative slope,
but it's

just
insignificantly
different from
zero.
But we do
find this
statistical
relationship
here.
So,
yeah.
and that's
what differentiates
cognitive
modeling from
purely
behavioral
work.
Okay.
Well,
let's see.
Next week,
we return.
Second discussion
on nine.
There are
some questions
already on nine.
And there's
probably also
a bunch more
we can add.
But we'll
come back to
nine next
week.
Any last
thoughts on

this?
Yeah.
You know,
maybe as an
extra thought,
maybe next week,
unless there are
any new
incomers who
do want
a refresher
as a summary
or something,
maybe going over
another example
or two
that you
could think
of.
I mean,
even this,
just at the very
end,
was super
helpful,
like the
correlation
graph between
cognitive
variables and
collected data.
So if you
had any
thoughts on
maybe something,
a simple
walkthrough
or two,

or maybe
opening that
up,
could be
really great.
Yeah,
just a direct
application and
the concrete
example is
super,
super helpful.
Cool.
Sounds good.
Yeah,
maybe if you
find a paper
that you
like,
then send
to me and
we can see
if it's
relevant or
maybe we
can look at
one of the
papers in
Table 9-1
or one or
two of
these high
saccade
papers.
Okay.
Sure.
Or we can
look at

another one.

But yeah.

Cool.

All right.

Thank you,

everybody.

Farewell.

Thank you so

much.

Bye.

Thank you.